Airport disruption in Germany

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Using only public EUROCONTROL and DWD data for 12 German airports, a LightGBM model ranked disrupted airport-days in 2024–2025 with average precision 0.39, against 0.24 for the previous 28-day disruption rate and a base rate of 0.06.

Time range: 2014–2025 · *Airports: 12* · *Airport-days: 52,596* · *Predictors: 37*

Models: logistic and ridge regression, LightGBM · *Validation: time-based split, test 2024–2025 evaluated once*

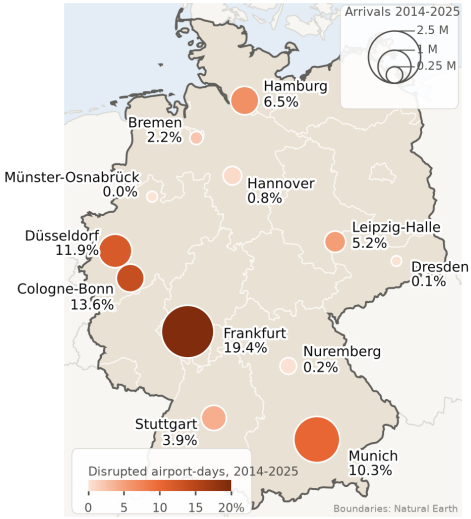


Fig. 1A The 12 airports. Bubble area: arrivals 2014–2025; colour and label: share of disrupted airport-days.

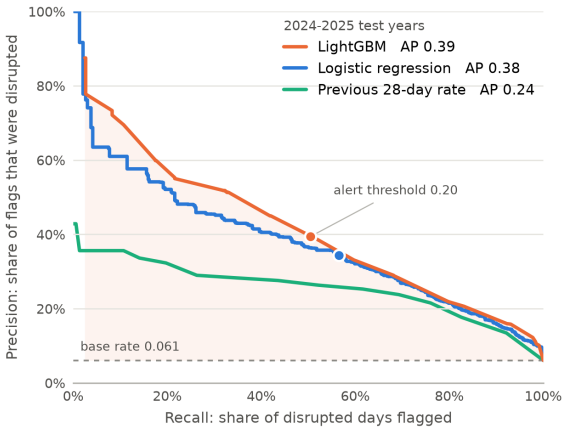


Fig. 1B Precision–recall curves (interpolated) on the 2024–2025 test years for LightGBM, logistic regression and the previous 28-day disruption rate. Dots: alert threshold 0.20 chosen on 2023; dashed line: base rate.

Background When an airport cannot accept all planned arrivals, EUROCONTROL's Network Manager holds flights at their departure airports. The resulting air traffic flow management (ATFM) delay is published per airport and day, with its cause [1, 2]. Weather is the largest recorded cause at German airports: 67% of all ATFM arrival delay minutes at the 12 airports studied here, 2014–2025. We asked whether public traffic and weather data are enough to anticipate which airport-days will be disrupted, defined as days on which at least 2% of arrivals receive more than 15 minutes of ATFM delay. Such days are concentrated at the large hubs: 19.4% of days at Frankfurt against 6.2% overall (Fig. 1A).

Results We joined daily EUROCONTROL arrival delays [Data 1] with hourly observations from the DWD station at each airport [Data 2] into 52,596 airport-days (Fig. 1E; all processing steps in [Protocol 1]). Models used 37 predictors: calendar, traffic and delay history from earlier days only, and same-day observed weather as a stand-in for a perfect forecast. They were trained on 2014–2021, tuned and calibrated on 2022, with the calibration method and alert threshold chosen on 2023, and scored once on 2024–2025 (8,771 airport-days, 533 disrupted).

LightGBM [4] reached an average precision (AP) of 0.39 and a ROC-AUC of 0.90; logistic regression [3] reached 0.38 and 0.89 (Fig. 1B, Table 1). At an alert threshold of 0.20 chosen on 2023, LightGBM flagged 682 days with a precision of 0.39 and a recall of 0.50. The gain over recent history was largest at Frankfurt (AP 0.63 vs 0.30) and smallest at Stuttgart, where both models caught only 3 of 52 disrupted days [Code 1].

The descriptive analysis points to weather and traffic. Thunderstorm days were 3.9 to 5.2 times as likely to be disrupted as other days at Frankfurt, Munich and Stuttgart (Fig. 1C). Disruption rose with traffic and with the number of adverse weather conditions, reaching 25.7% on days with two conditions and 100–120% of the airport's median traffic (Fig. 1D). Predicting delay minutes per arrival was harder: LightGBM barely beat predicting zero (MAE 0.200 vs 0.207; Table 1). Because observed weather replaces a forecast, all scores are an upper bound for day-ahead use.

Disrupted day?	AP	ROC-AUC	Brier	Prec.	Recall
LightGBM classifier	**0.389**	**0.895**	**0.045**	**0.39**	0.50
Logistic regression	0.376	0.889	0.046	0.34	0.56
28-day rate	0.242	0.848	0.050	0.25	**0.62**
Constant rate	0.061	0.500	0.057	–	–

Delay min per arrival	MAE	RMSE	R^2
LightGBM (Tweedie)	**0.200**	**0.947**	**0.16**
Ridge regression	0.290	0.957	0.14
Always zero	0.207	1.054	−0.04
28-day average	0.298	0.989	0.08

Table 1 Test results on 2024–2025 (8,771 airport-days, scored once), best value per column in bold. Left: classifiers, with precision and recall at the alert threshold chosen on 2023 (0.20; 0.15 for the 28-day rule). Right: regressors. Lower is better for Brier, MAE and RMSE.

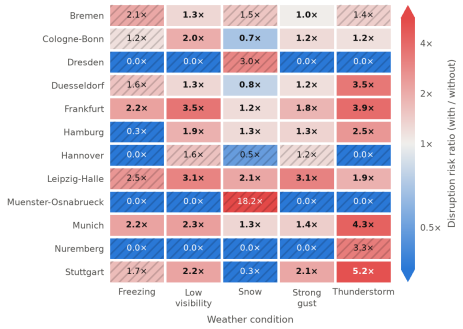


Fig. 1C Disruption rate on days with a weather condition divided by the rate on days without it, 2014–2025; a missing reading counts as no event. Hatched: fewer than 10 disrupted days with the condition.

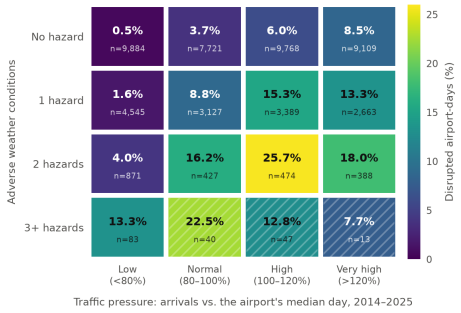


Fig. 1D Disruption rate by traffic pressure (arrivals relative to the airport's median day, 2014–2025) and weather severity (number of adverse conditions). n = days per cell; hatched: fewer than 50 days.

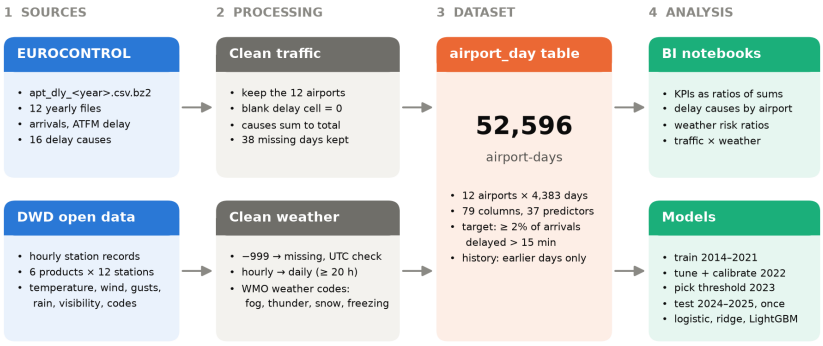


Fig. 1E From the two public sources to the airport-day table used by the BI notebooks and the models; every step is documented in [Protocol 1].

References

[1] EUROCONTROL Performance Review Unit. ATFM delay definition. https://ansperformance.eu/definition/atfm-delay/

[2] EUROCONTROL Performance Review Unit. Airport arrival ATFM delays. https://ansperformance.eu/capacity/atfm_apt

[3] Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research* 12, 2825–2830. https://jmlr.org/papers/v12/pedregosa11a.html

[4] Ke, G. et al. (2017). LightGBM: A Highly Efficient Gradient Boosting Decision Tree. *Advances in Neural Information Processing Systems* 30. https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html

Code

[1] Project repository: data pipeline, BI and model notebooks, figure script. https://github.com/sidnov6/german-airport-disruption

Data

[1] EUROCONTROL. Daily airport arrival ATFM delay files `apt_dly_YYYY.csv.bz2`, 2014–2025. https://www.eurocontrol.int/performance/data/download/csv/apt_dly_2025.csv.bz2 (one file per year)

[2] Deutscher Wetterdienst, Climate Data Center. Hourly station observations (CC BY 4.0). https://opendata.dwd.de/climate_environment/CDC/observations_germany/climate/hourly/

[3] Natural Earth 1:10m admin boundaries (public domain), used for the map in Fig. 1A. https://www.naturalearthdata.com

[4] Processed dataset (52,596 airport-days, 79 columns) with feature contract and quality report. https://github.com/sidnov6/german-airport-disruption/tree/main/data/processed

Protocols

[1] Data sources and processing: download, cleaning, hourly-to-daily aggregation, targets and features. https://github.com/sidnov6/german-airport-disruption/blob/main/docs/DATA.md

[2] Model training and evaluation protocol (time-based split, tuning, calibration, single test evaluation). https://github.com/sidnov6/german-airport-disruption/tree/main/notebooks

Use of AI tools We used a large language model, Claude (Anthropic), as an assistant in three parts of this project: writing and reviewing code for the data pipeline, notebooks and figures; building the presentation slides; and drafting and typesetting this micropublication. The authors designed the study, reviewed all code, results and text, and take full responsibility for the content. Every reported number can be traced to the saved outputs in [Code 1].